



The Scots College

HSC Mathematics Extension 2

Trial

August 2010

Name: _____

General Instructions

- Working time : 3 hours + 5 minutes
Reading time.
- Write using blue or black pen
- Board approved calculators may be
Used (Non Graphic)
- All necessary working should be
shown in every question
- Standard Integrals Table attached

TOTAL MARKS: 120

Attempt Questions 1 - 8
All questions are of equal value

WEIGHTING: 40 %

Answer each question in a SEPARATE writing booklet.

Question 1 (Marks 15) Use a SEPARATE writing booklet.

- a) Evaluate [2]

$$\int_0^1 \frac{2 \, dx}{\sqrt{2-x^2}}$$

- b) Find [2]

$$\int \frac{\tan x}{\ln(\cos x)} \, dx$$

- c) Evaluate [3]

$$\int_0^1 \sin^{-1} x \, dx$$

- d) [4]

- i) Find the real numbers a , b , and c such that

$$\frac{1+4x}{(x^2+1)(4-x)} \equiv \frac{a}{4-x} + \frac{bx+c}{x^2+1}$$

- ii) Hence evaluate

$$\int_0^2 \frac{1+4x}{(x^2+1)(4-x)} \, dx$$

- e) By using the substitution $x = 2 \cos \theta$ or otherwise evaluate [4]

$$\int_{-2}^0 \sqrt{4-x^2} \, dx$$

Question 2 (Marks 15) Use a SEPARATE writing booklet.

a) Express $\left(\frac{1+i}{1-i}\right)^3$ in the form $a + ib$; where a and b are real numbers [2]

b) i) Write $\frac{5-i}{2-3i}$ in the form $x + iy$, where x and y are real numbers. [2]

ii) Write $\frac{5-i}{2-3i}$ in the modulus-argument form. [3]

iii) Calculate $\left(\frac{5-i}{2-3i}\right)^4$. [1]

c) Sketch the region on the Argand diagram where the inequalities [3]

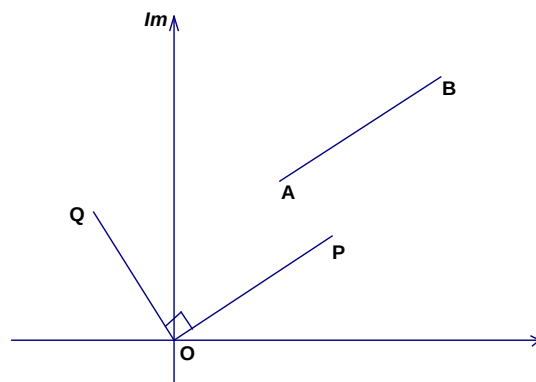
$$|z - \bar{z}| < 4 \quad \text{and} \quad |z + 1| > 1$$

hold simultaneously.

d) In the Argand diagram below, $AB = OP = OQ$, $OP \parallel AB$ and $OP \perp OQ$. [4]
A represents the complex number $5 + 3i$ and B represents $11 + 5i$. Copy this diagram into your answer scripts and find the complex numbers represented by the point :

i) P

ii) Q



Question 3 (Marks 15) Use a SEPARATE writing booklet.

- a) i) Determine whether $f(x) = \frac{x}{x^2-1}$ is an odd or even function. [6]

ii) Sketch the graph of $y = f(x)$.

iii) Using the graph of $y = f(x)$, sketch on separate axes, the graphs of

(α) $y = f(-x)$

(β) $y = |f(x)|$

(γ) $y = [f(x)]^2$

- b) Tangents to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at points $P(x_1, y_1)$ and $Q(x_2, y_2)$ intersect at T . M is the mid-point of PQ . [6]

i) Given that the tangent to the ellipse at P has equation $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$ write down the equation of the tangent to the ellipse at Q .

ii) Show that the line $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = \frac{xx_2}{a^2} + \frac{yy_2}{b^2}$ passes through T and M .

iii) Deduce that O, T and M are collinear.

iv) If $\angle PTQ$ is a right angle, show that $\frac{x_1x_2}{a^4} + \frac{y_1y_2}{b^4} = 0$

- c) i) If $a > b > 0$, sketch the curve and shade the region $\int_b^a \sqrt{a^2 - x^2} dx$. [3]

ii) By using your diagram, or otherwise, show that

$$\int_b^a \sqrt{a^2 - x^2} dx = \frac{a^2}{2} \cos^{-1} \left(\frac{b}{a} \right) - \frac{b}{a} \sqrt{a^2 - x^2}$$

Question 4 (Marks 15) Use a SEPARATE writing booklet.

- a) i) Sketch the graph of the curve $y = x + e^{-x}$ showing clearly the coordinates of any turning points and the equations of any asymptotes. [2]

- ii) The region in the first quadrant between the curve $y = x + e^{-x}$ and the line $y = x$ and bounded by the line $x = 1$ is rotated through one complete revolution about the y-axis. Use the method of cylindrical shells to show that the volume V of the solid of revolution is given by [3]

$$V = 2\pi \int_0^1 x e^{-x} dx$$

- iii) Find the volume of the solid of revolution. [3]

- b) Let $f(x) = x^2(x^2 - 2)$. The tangent to the curve $y = f(x)$ at the point A with x coordinate α meets the curve again at B . [7]

- i) Show the tangent AB has equation $y = 4\alpha(\alpha^2 - 1)x + \alpha^2(2 - 3\alpha^2)$.

- ii) Deduce that $x^2(x^2 - 2) = 4\alpha(\alpha^2 - 1)x + \alpha^2(2 - 3\alpha^2)$ has real roots $\alpha, \alpha, \beta, \gamma$ for some β, γ .

- iii) For $\alpha \neq 0$, find $\beta + \gamma$ and $\beta\gamma$ in terms of α and write down a quadratic equation with roots β, γ .

- iv) Find the possible values of α .

Question 5 (Marks 15) Use a SEPARATE writing booklet.

a) If $u_1 = 12$, $u_2 = 30$ and $u_n = 5u_{n-1} - 6u_{n-2}$ for $n \geq 3$. [5]

(i) Determine u_3 and u_4 .

(ii) Show that $u_n = 2 \times 3^n + 3 \times 2^n$ for $n = 1$ **and** $n = 2$.

(iii) If $u_k = 2 \times 3^k + 3 \times 2^k$ and $u_{k+1} = 2 \times 3^{k+1} + 3 \times 2^{k+1}$,
where k is a positive integer, prove that
$$u_{k+2} = 2 \times 3^{k+2} + 3 \times 2^{k+2}.$$

b) (i) Show that $\operatorname{cosec} 2\theta + \cot 2\theta = \cot \theta$ for all real values of θ . [7]

(ii) Use the result above to :

(α) find in surd form the values of $\cot \frac{\pi}{8}$ and $\cot \frac{\pi}{12}$.

(β) show without using calculators that

$$\operatorname{cosec} \frac{4\pi}{15} + \operatorname{cosec} \frac{8\pi}{15} + \operatorname{cosec} \frac{16\pi}{15} + \operatorname{cosec} \frac{32\pi}{15} = 0$$

c) i) Show that for $a > 0$ and $n \neq 0$, [3]

$$\log_{a^n}(x) = \frac{1}{n} \log_a x.$$

ii) Hence evaluate

$$\log_2 3 + \log_4 3 + \log_{16} 3 + \log_{256} 3 + \cdots.$$

Question 6 (Marks 15) Use a SEPARATE writing booklet.

- a) The base of a certain solid is the region between the x-axis and the curve $y = \sin 2x$ between $x = 0$ and $x = \frac{3\pi}{8}$. [6]

Each plane section of the solid perpendicular to the x-axis is an equilateral triangle with one side in the base of the solid.

Find the volume of the solid.

- b) In the expansion of $(ax - bx^{-2})^8$ the coefficient of x^2 and x^{-1} are equal. [4]

Show that $a + 2b = 0$.

- c) A particle of mass m kg falls from rest in a medium where the resistance to motion is mkv when the particle has velocity v ms^{-1} . [5]

i) Draw a diagram showing the forces acting on the particle.

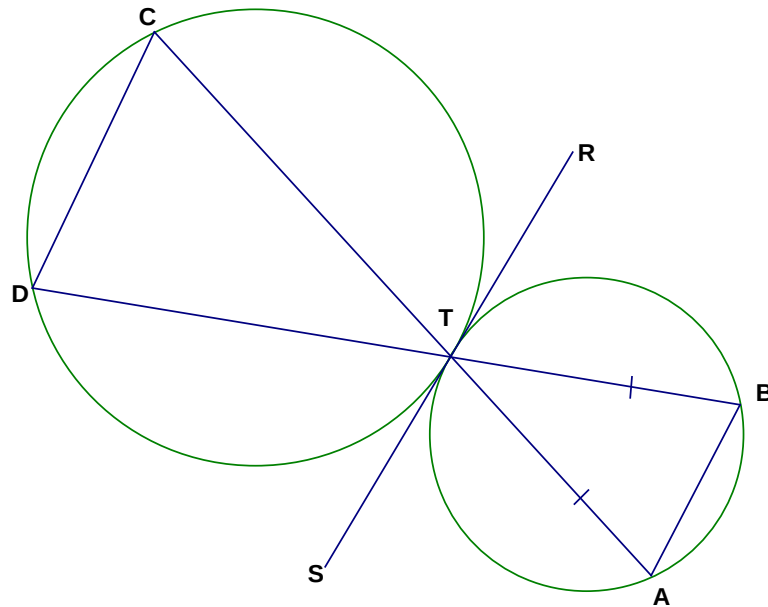
ii) Show that the equation of motion of the particle is $\ddot{x} = k(V - v)$ where V ms^{-1} is the terminal velocity of the particle in this medium, and x metres is the distance fallen in t seconds.

iii) Find the time T seconds taken for the particle to attain 50% of its terminal **velocity**, and the **distance** fallen in time in terms of V and k .

Question 7 (Marks 15) Use a SEPARATE writing booklet.

- a) Two circles touch externally at a point T . [8]

A and B are points on the first circle such that $AT = BT$, and AT and BT produced meet the second circle at C and D respectively. RS is the common tangent at T . Let $\angle BAT = \alpha$.



- i) Copy the diagram and include the information above.

- ii) Prove that $\angle BAC = \angle ACD$.

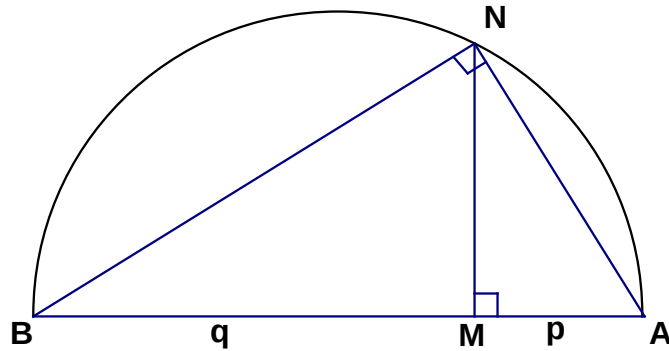
- iii) Prove that $ABCD$ is a trapezium with two equal sides.

The line BC cuts the first circle in V and the second circle again in W , and the line AD cuts the first circle in U and the second circle again in X .

- iv) Prove that the points U , V , W and X are concyclic.

Question 7 continued.....

- b) In the diagram, AB is the diameter of a semicircle. The angle ANB is 90° [7]
and M is a point on AB such that NM is perpendicular to AB .



If $AM = p$ and $MB = q$,

i) Show that $NM = \sqrt{pq}$.

ii) Deduce, using the diagram, that $\sqrt{pq} \leq \frac{p+q}{2}$.

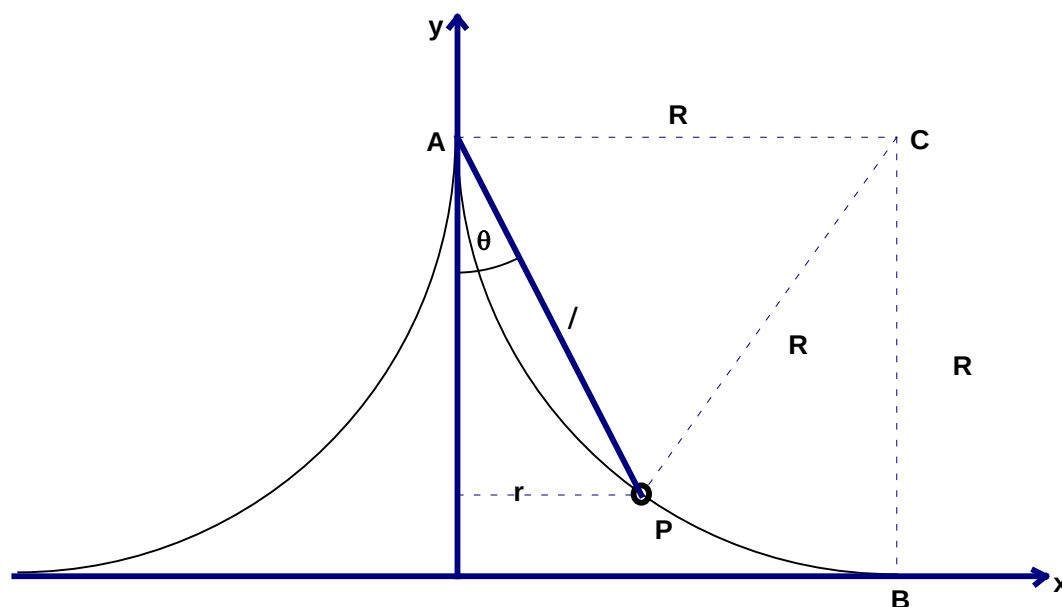
iii) Use (ii) to prove that if $p, q, x, y > 0$, then

$$\frac{1}{4}(p + q + x + y) \geq (pqxy)^{\frac{1}{4}}$$

iv) Deduce that if $l, m, n, z > 0$, then $\frac{l}{m} + \frac{m}{n} + \frac{n}{z} + \frac{z}{l} \geq 4$

End of Question 7

[10]



vi) Deduce that there is a maximum value ω for the motion to occur as described, and write down this maximum value.

Question 8 continued.....

- b) i) Show that $\tan^{-1}(n+1) - \tan^{-1}(n-1) = \tan^{-1} \frac{2}{n^2}$, where n is a positive integer. [5]

ii) Hence or otherwise show that for $n \geq 1$

$$\sum_{r=1}^n \tan^{-1} \frac{2}{r^2} = \frac{3\pi}{4} + \tan^{-1} \frac{2n+1}{1-n-n^2}$$

End of Assessment

Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Q1

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$$\begin{aligned}
 Q1(a) \quad \int_0^1 \frac{2}{\sqrt{2-x^2}} dx &= 2 \int_0^1 \frac{1}{\sqrt{2-x^2}} dx \\
 &= 2 \left[\sin^{-1} \frac{x}{\sqrt{2}} \right]_0^1 \\
 &= 2 \left[\sin^{-1} \frac{1}{\sqrt{2}} - \sin 0 \right] \\
 &= 2 \left(\frac{\pi}{4} \right) \\
 &= \frac{\pi}{2}
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad \int \frac{\tan x}{\ln(\cos x)} dx & \quad \text{Let } u = \ln(\cos x) \\
 \frac{du}{dx} &= \frac{1}{\cos x} (-\sin x) \\
 du &= -\tan x dx \\
 -\frac{du}{\tan x} &= dx
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow \int \frac{\tan x}{u} \cdot \frac{-du}{\tan x} & \\
 - \int \frac{du}{u} & \\
 - \ln(\ln(\cos x)) + C &
 \end{aligned}$$

$$\begin{aligned}
 (c) \quad \int_0^1 \sin^{-1} x dx &= \int_0^1 1 \cdot \sin^{-1} x dx \\
 &= (\sin^{-1} x) \cdot x - \int x \frac{1}{\sqrt{1-x^2}} dx \\
 &= x \sin^{-1} x \Big|_0^1 + \frac{1}{2} \int (1-x^2)^{-1/2} (-2x) dx \\
 &= x \sin^{-1} x \Big|_0^1 + \frac{1}{2} x \frac{1}{1/2} (1-x^2)^{1/2} \Big|_0^1 \\
 &= \left[x \sin^{-1} x + \sqrt{1-x^2} \right]_0^1 \\
 &= \left[(1 \cdot \sin^{-1}(1) + 0) - (0 + \sqrt{1}) \right] \\
 &= \frac{\pi}{2} - 1 \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 (d) \quad \frac{1+4x}{(x^2+1)(4-x)} &= \frac{A}{4-x} + \frac{Bx+C}{x^2+1} \\
 1+4x &= A(x^2+1) + (Bx+C)(4-x) \\
 x=0 \quad A+4C &= 1 \quad \text{--- (1)} \\
 x=4 \quad 17A &= 17 \quad \text{--- (2)} \\
 A &= 1 \\
 x=1 \quad 2A+3B+C &= 5 \quad \text{--- (3)} \\
 \text{from (1) \& (2)} \quad C &= 0 \\
 \text{from (3)} \quad B &= 1
 \end{aligned}$$

$$\begin{aligned}
 \therefore \int_0^2 \frac{(1+4x)}{(4-x)(x^2+1)} dx &= \int_0^2 \left[\frac{1}{4-x} + \frac{x}{x^2+1} \right] dx \\
 &= \left[-\ln(4-x) + \frac{1}{2} \ln(x^2+1) \right]_0^2 \\
 &= \left[\ln \frac{\sqrt{x^2+1}}{(4-x)} \right]_0^2 \\
 &= \ln \frac{\sqrt{5}}{2} - \ln \frac{1}{4} \\
 &= \ln(2\sqrt{5})
 \end{aligned}$$

$$\begin{aligned}
 (e) \quad x &= 2 \cos \theta \\
 dx &= -2 \sin \theta d\theta \\
 x=-2 \quad \theta &= \pi \\
 x=0 \quad \theta &= \pi/2 \\
 \int_{\pi}^{\pi/2} \sqrt{4-4\cos^2 \theta} \cdot (-2 \sin \theta) d\theta &\quad \checkmark \\
 &= -2 \int_{\pi}^{\pi/2} 2 \sin \theta \sin \theta d\theta \\
 &= -2 \int_{\pi}^{\pi/2} 2 \sin^2 \theta d\theta \\
 &= -2 \int_{\pi}^{\pi/2} (1 - \cos 2\theta) d\theta \quad \checkmark \\
 &= -2 \left[(\pi/2 - 0) - (\pi - 0) \right] \\
 &= \pi \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \text{Q2(a) consider } \frac{1+i}{1-i} &= \frac{1+i}{1-i} \times \frac{1+i}{1+i} \\
 &= \frac{1+2i+i^2}{1-i^2} \\
 &= \frac{1+2i-1}{2} \\
 &= i = 0+1i
 \end{aligned}$$

$$\therefore \left(\frac{1+i}{1-i}\right)^3 = i^3 = i \cdot i^2 = -i = 0 - 1i$$

$$\begin{aligned}
 \text{(b) i) } \frac{5-i}{2-3i} &= \frac{5-i}{2-3i} \times \frac{2+3i}{2+3i} \\
 &= \frac{10+15i-2i+3}{2^2-(3i)^2} = \frac{13+13i}{4+9} \\
 &= 1+i
 \end{aligned}$$

$$\text{(ii) } \sqrt{1^2+1^2} = \sqrt{2}$$

$$\text{let } 1+i = r(\cos \theta + i \sin \theta)$$

$$r \cos \theta = 1 \quad r \sin \theta = 1$$

$$\tan \theta = 1$$

$$\theta = 45^\circ, 225^\circ$$

Since $\cos \theta$ and $\sin \theta$ are positive, θ lies in 1st Quad.

$$\text{Hence } \theta = 45^\circ$$

$$\frac{5-i}{2-3i} = \sqrt{2} (\cos 45^\circ + i \sin 45^\circ) \checkmark$$

$$\begin{aligned}
 \text{(iii) } \left(\frac{5-i}{2-3i}\right)^4 &= \left(\sqrt{2} \left(\cos \frac{\pi}{4}\right)\right)^4 \\
 &= 4 \cos \pi \quad \checkmark
 \end{aligned}$$

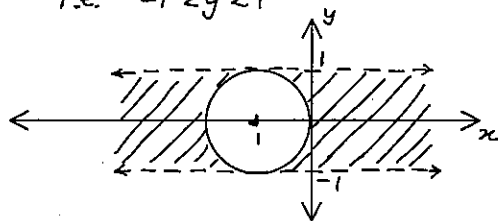
$$\text{(c) } |z-\bar{z}| < 2 \text{ and } |z-1| \geq 1$$

$$z = x+iy, \bar{z} = x-iy$$

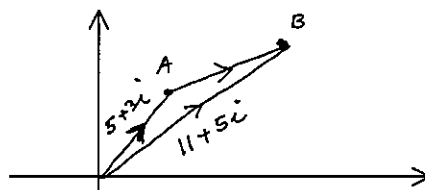
$$\therefore z-\bar{z} = 2iy$$

$$\begin{aligned}
 \text{hence } |z-\bar{z}| &= |2iy| \\
 &= |2y| < 2 \\
 &\Rightarrow |y| < 1
 \end{aligned}$$

$$\text{i.e. } -1 < y < 1$$



(d)



$$\begin{aligned}
 \vec{AB} &= (11+5i) - (5+3i) \\
 &= 6+2i \quad (2)
 \end{aligned}$$

$$P = \vec{AB} = 6+2i$$

$$\begin{aligned}
 Q &= i \times P \\
 &= i(6+2i) \\
 &= -2+6i \quad (2)
 \end{aligned}$$

Q3 a

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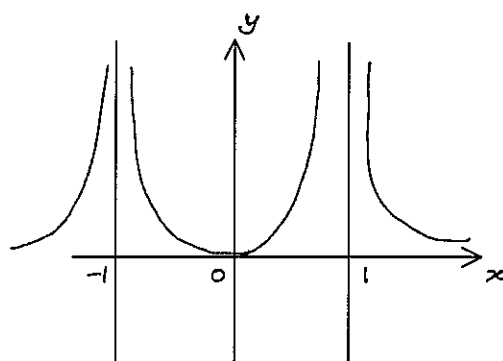
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$$3(a) i) \quad f(x) = \frac{x}{x^2-1}$$

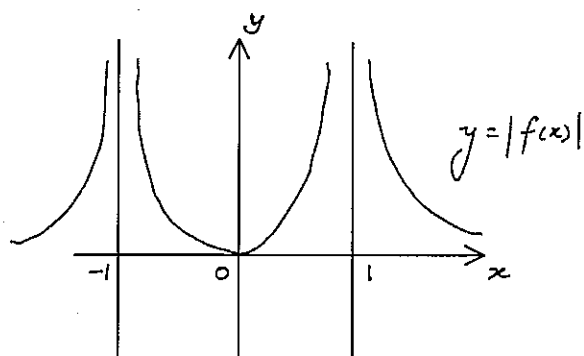
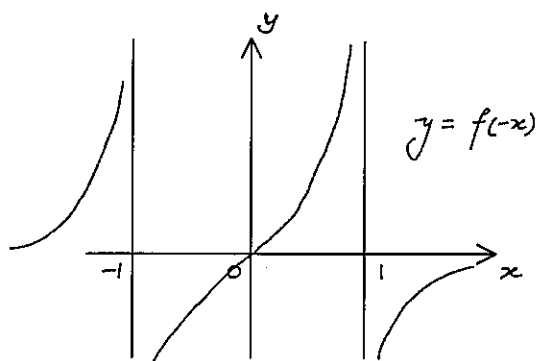
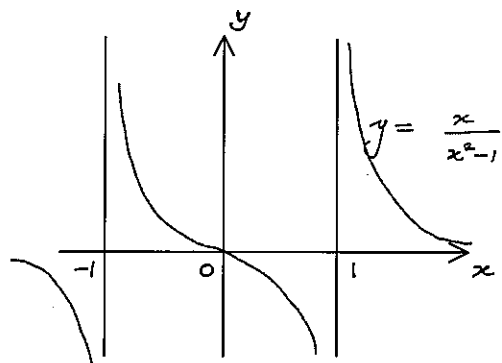
$$f(-x) = \frac{-x}{x^2-1}$$

$$= -f(x)$$

$\therefore$ odd function



(ii)



Q3 b

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$$3(b)i) \text{ Tangent at } P: \frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1 \quad (1)$$

$$" \quad Q: \frac{xx_2}{a^2} + \frac{yy_2}{b^2} = 1 \quad (2)$$

(ii) T lies on both tangents

$$\therefore \frac{xx_1}{a^2} + \frac{yy_1}{b^2} = \frac{xx_2}{a^2} + \frac{yy_2}{b^2} \quad (3)$$

$$\Rightarrow \frac{x(x_1 - x_2)}{a^2} + \frac{y(y_1 - y_2)}{b^2} = 0 \quad (4)$$

$$\text{at } M\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

$$\begin{aligned} (4) \Rightarrow L.H.S. &= \frac{(x_1 + x_2)(x_1 - x_2)}{2a^2} + \frac{(y_1 + y_2)(y_1 - y_2)}{2b^2} \\ &= \frac{1}{2} \left\{ \left(\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} \right) - \left(\frac{x_2^2}{a^2} + \frac{y_2^2}{b^2} \right) \right\} \end{aligned}$$

$$= \frac{1}{2} \{1 - 1\}$$

$$= 0$$

i.e. coordinates of M also satisfy (3)

iii) (3) is the equation of line TM and $(0,0)$ satisfies this eqn.
Hence O, T, M are collinear

(iv) using (4) of eqn TM

$$m_1 \text{ of } TM \text{ is } -\frac{b^2(x_1 - x_2)}{a^2(y_1 - y_2)}$$

$$m_2 \text{ of } PQ \text{ is } \frac{(y_1 - y_2)}{(x_1 - x_2)}$$

$$\text{Product } m_1 m_2 = -\frac{b^2}{a^2}$$

$$PT \text{ has gradient } m_3 = -\frac{b^2 x_1}{a^2 y_1}$$

$$QT \text{ has gradient } m_4 = -\frac{b^2 x_2}{a^2 y_2}$$

$$PT \perp QT \Rightarrow m_3 m_4 = -1$$

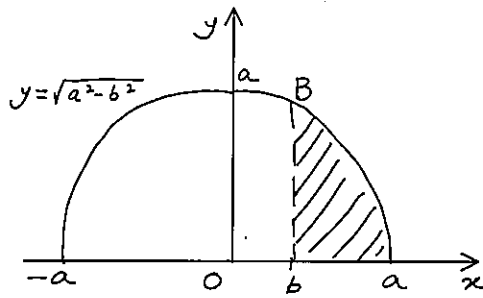
$$\therefore \frac{b^4 x_1 x_2}{a^4 y_1 y_2} = -1$$

$$\therefore \frac{x_1 x_2}{a^4} + \frac{y_1 y_2}{b^4} = 0$$

Q3 c

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3c i) $a > b > 0$, $y = \sqrt{a^2 - x^2}$



ii) $\int_b^a \sqrt{a^2 - x^2} dx = \text{sector OBA} - \Delta OBC$
 $\cos \angle BOC = \frac{b}{a} \Rightarrow \angle BOC = \cos^{-1}\left(\frac{b}{a}\right)$

Area of sector BOA = $\frac{1}{2} a^2 \cos^{-1}\left(\frac{b}{a}\right)$

$BC = \sqrt{a^2 - b^2}$

Area $\Delta BOC = \frac{b}{2} \sqrt{a^2 - b^2}$

$\therefore \int_b^a \sqrt{a^2 - x^2} dx = \frac{a^2}{2} \cos^{-1}\left(\frac{b}{a}\right) - \frac{b}{2} \sqrt{a^2 - b^2}$

OR

Let $x = a \cos \theta \Rightarrow \frac{dx}{d\theta} = -a \sin \theta$

$\int_b^a \sqrt{a^2 - x^2} dx = \int \sqrt{a^2 - a^2 \cos^2 \theta} (-a \sin \theta d\theta)$
 $= -a^2 \int \sin^2 \theta d\theta$

$= -a^2 \int \frac{1 - \cos 2\theta}{2} d\theta$

$= -\frac{a^2}{2} \left(\theta - \frac{\sin 2\theta}{2} \right) + C$

$= -\frac{a^2}{2} \left(\theta - \sin \theta \cos \theta \right) + C$

$\cos \theta = \frac{x}{a}$, $\sin \theta = \frac{\sqrt{a^2 - x^2}}{a}$, $\theta = \cos^{-1}\left(\frac{x}{a}\right)$

$= -\frac{a^2}{2} \left[\cos^{-1}\left(\frac{x}{a}\right) - \frac{x \sqrt{a^2 - x^2}}{a^2} \right]_b^a$

$= \frac{a^2}{2} \left[-\cos^{-1}\left(\frac{a}{a}\right) + 0 - \left\{ -\cos^{-1}\left(\frac{b}{a}\right) + \frac{b \sqrt{a^2 - b^2}}{a^2} \right\} \right]$

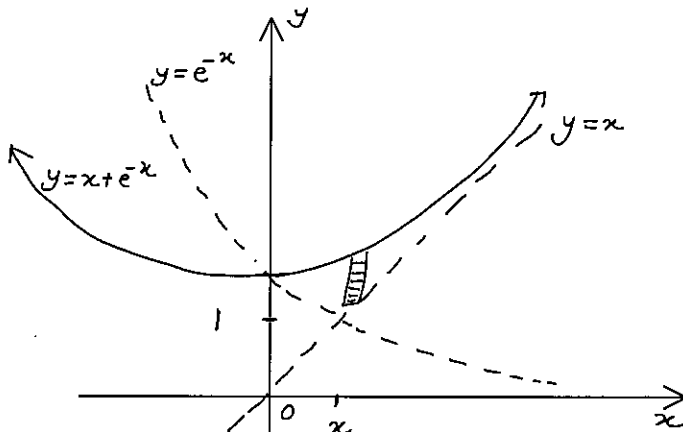
$= \frac{a^2}{2} \left[0 + \cos^{-1}\left(\frac{b}{a}\right) - \frac{b \sqrt{a^2 - b^2}}{a^2} \right]$

$= \frac{a^2}{2} \cos^{-1}\left(\frac{b}{a}\right) - \frac{b \sqrt{a^2 - b^2}}{2}$

Q4 a

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(i)



$$y' = 1 - e^{-x} = 0$$

$$e^{-x} = 1$$

$x = 0$ at Turning Pt.

$(0, 1)$

$$y'' = e^{-x} = 1 > 0 \therefore \text{min}$$

ii)

$$V = 2\pi \int r h dx$$

$$= 2\pi \int_0^1 x(x + e^{-x} - x) dx$$

$$= 2\pi \int_0^1 x e^{-x} dx \quad \checkmark$$

$$\text{iii)} \quad V = 2\pi \int_0^1 x \frac{d}{dx} (-e^{-x}) dx$$

$$= 2\pi \left[(-x e^{-x}) \Big|_0^1 + \int_0^1 e^{-x} dx \right]$$

$$= 2\pi \left[-e^{-1} + [-e^{-x}]_0^1 \right]$$

$$= 2\pi \left[-e^{-1} + (-e^{-1} + 1) \right]$$

$$V = 2\pi \left(1 - \frac{2}{e} \right) \quad u^3$$

Q4 b

Wednesday, 11 August 2010
8:37 PM

4b i) $f(x) = x^2(x^2 - 2)$

$\Rightarrow f'(x) = 4x(x^2 - 1)$

$\therefore$ Gradient of Tangent

at $A(\alpha, f(\alpha))$ is $4\alpha(\alpha^2 - 1)$

Equation of Tangent

$y - \alpha^2(\alpha^2 - 2) = 4\alpha(\alpha^2 - 1)(x - \alpha)$

$y = 4\alpha(\alpha^2 - 1)x + \alpha^2(2 - 3\alpha^2)$

(ii) The tangent meets the curve where

$x^2(x^2 - 2) = 4\alpha(\alpha^2 - 1)x + \alpha^2(2 - 3\alpha^2) \quad \text{--- (1)}$

Tangent touches at A

$\Rightarrow \alpha$ is a double root of (1)

Since Tangent meets curve again at B, equation (1) has another real root β , where β is the x coordinate of B.

Since equation (1) has real coeffs., the fourth root γ cannot be non-real as such roots come in complex conjugate pairs.

Hence eqn (1) has real roots $\alpha, \alpha, \beta, \gamma$.

iii) Rearranging (1) we get:

$x^4 - 2x^2 - 4\alpha(\alpha^2 - 1)x + \alpha^2(3\alpha^2 - 2) = 0 \quad \text{--- (2)}$

For $\alpha \neq 0$

$2\alpha + \beta + \gamma = 0 \Rightarrow \beta + \gamma = -2\alpha$

$\alpha^2\beta\gamma = \alpha^2(3\alpha^2 - 2) \Rightarrow \beta\gamma = (3\alpha^2 - 2)$

Hence β & γ are the roots of the quadratic equation

$x^2 + 2\alpha x + (3\alpha^2 - 2) = 0 \quad \text{--- (3)}$

(iv) For $\alpha \neq 0$, since (3) has real roots β, γ

$\Delta = 4\alpha^2 - 4(3\alpha^2 - 2)$

$= 8(1 - \alpha^2) \geq 0$ and $\alpha^2 \leq 1$

For $\alpha = 0$, tangent at A is the x-axis, meeting the curve again in $(\sqrt{2}, 0), (-\sqrt{2}, 0)$.

Hence $-1 \leq \alpha \leq 1$

Q5 a

Wednesday, 18 August 2010

7:48 PM

$$\begin{aligned} \text{Q5 a i)} \quad u_3 &= 5(30) - 6(12) = 78 \\ u_4 &= 5(78) - 6(30) = 210 \end{aligned} \quad \left. \vphantom{\begin{aligned} u_3 &= 5(30) - 6(12) = 78 \\ u_4 &= 5(78) - 6(30) = 210 \end{aligned}} \right\} \checkmark$$

$$\text{ii)} \quad n=1,$$

$$\begin{aligned} 2 \cdot 3^n + 3 \cdot 2^n &= 2 \cdot 3^1 + 3 \cdot 2^1 \\ &= 6 + 6 = 12 = u_1 \end{aligned}$$

$$n=2,$$

$$\begin{aligned} 2 \cdot 3^n + 3 \cdot 2^n &= 2 \cdot 3^2 + 3 \cdot 2^2 \\ &= 18 + 12 = 30 = u_2 \end{aligned}$$

$$\text{So } u_n = 2 \cdot 3^n + 3 \cdot 2^n \text{ for } n=1, 2 \quad \checkmark$$

$$\begin{aligned} \text{iii)} \quad u_{k+2} &= 5u_{k+1} - 6u_k \\ &= 5(2 \cdot 3^{k+1} + 3 \cdot 2^{k+1}) - 6(2 \cdot 3^k + 3 \cdot 2^k) \\ &= 10 \cdot 3^{k+1} + 15 \cdot 2^{k+1} - 12 \cdot 3^k - 18 \cdot 2^k \\ &= 10 \cdot 3^{k+1} + 15 \cdot 2^{k+1} - 4 \cdot 3^{k+1} - 9 \cdot 2^{k+1} \\ &= 6 \cdot 3^{k+1} + 6 \cdot 2^{k+1} \\ &= 2 \cdot 3^{k+2} + 3 \cdot 2^{k+2} \quad \checkmark \end{aligned}$$

Q5 b

Wednesday, 11 August 2010
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$$\begin{aligned}
 Q5b i) \quad \operatorname{cosec} 2\theta + \cot 2\theta &= \frac{1}{\sin 2\theta} + \frac{\cos 2\theta}{\sin 2\theta} \\
 &= \frac{1 + \cos^2 \theta - \sin^2 \theta}{2 \sin \theta \cos \theta} \\
 &= \frac{2 \cos^2 \theta}{2 \sin \theta \cos \theta} \\
 &= \cot \theta
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad \cot \frac{\pi}{8} &= \operatorname{cosec} \frac{\pi}{4} + \cot \frac{\pi}{4} \\
 &= \sqrt{2} + 1 \\
 \cot \frac{\pi}{12} &= \operatorname{cosec} \frac{\pi}{6} + \cot \frac{\pi}{6} \\
 &= 2 + \sqrt{3}
 \end{aligned}$$

$$\begin{aligned}
 (iii) \quad & \operatorname{cosec} \frac{4\pi}{15} + \operatorname{cosec} \frac{8\pi}{15} + \operatorname{cosec} \frac{16\pi}{15} + \operatorname{cosec} \frac{32\pi}{15} \\
 &= \left[\cot \frac{2\pi}{15} - \cot \frac{4\pi}{15} \right] + \left[\cot \frac{4\pi}{15} - \cot \frac{8\pi}{15} \right] \\
 &\quad + \left[\cot \frac{8\pi}{15} - \cot \frac{16\pi}{15} \right] + \left[\cot \frac{16\pi}{15} - \cot \frac{32\pi}{15} \right] \\
 &= \cot \frac{2\pi}{15} - \cot \frac{32\pi}{15} \quad \checkmark \\
 &= \cot \frac{2\pi}{15} - \cot \left[2\pi + \frac{2\pi}{15} \right] \\
 &= \cot \frac{2\pi}{15} - \cot \frac{2\pi}{15} \\
 &= 0 \quad \checkmark
 \end{aligned}$$

5(xv)

$$\begin{aligned}
 \log_{a^n} x &= \frac{\log_a x}{\log_n a^n} \\
 &= \frac{\log_a x}{n}
 \end{aligned}$$

$$ii) \quad \log_{2^n} 3 = \frac{1}{n} \log_2 3$$

$$\Rightarrow \log_2 3 + \log_4 3 + \log_8 3 + \log_{16} 3 + \dots$$

$$\Rightarrow \left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots \right) \log_2 3$$

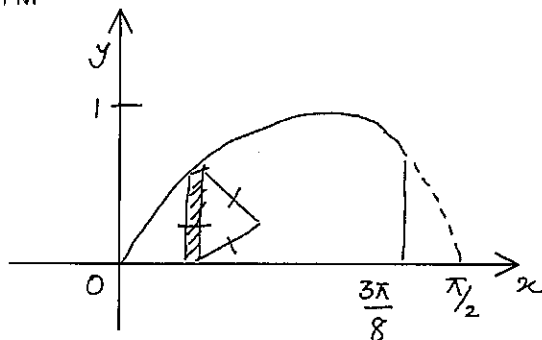
$$\Rightarrow 2 \log_2 3$$

Q6 a

Tuesday, 10 August 2010

8:22 PM

6(a)



$$\text{Vol. of Sphere} = \frac{1}{2} \cdot y \cdot y \cdot \sin 60^\circ \cdot \delta x$$

$$\therefore V = \frac{\sqrt{3}}{4} \int_0^{3\pi/8} y^2 dx$$

$$= \frac{\sqrt{3}}{4} \int_0^{3\pi/8} \sin^2 2x dx$$

$$= \frac{\sqrt{3}}{8} \int_0^{3\pi/8} 1 - \cos 4x dx$$

$$= \frac{\sqrt{3}}{8} \left[x - \frac{\sin 4x}{4} \right]_0^{3\pi/8}$$

$$= \frac{\sqrt{3}}{8} \left[\frac{3\pi}{8} - \frac{-1}{4} - 0 \right]$$

$$= \frac{\sqrt{3} (3\pi + 2)}{8}$$

$$\approx 0.31 \pi^3$$

$$6(b) (ax - bx^{-2})^8$$

$$T_{k+1} = 8C_k (ax)^{8-k} (-bx^{-2})^k$$

$$\therefore 8-k-2k=2$$

$$6=3k$$

$$k=2$$

$$\text{or } 8-3k=-1$$

$$9=3k$$

$$k=3$$

$$\therefore \text{Coeff. of } T_3 = \text{Coeff. of } T_4$$

$$8C_2 a^6 (-b)^2 = 8C_3 a^5 (-b)^3$$

$$\frac{8 \times 7}{2} a^6 b^2 = \frac{8 \times 7 \times 6}{3 \times 2} a^5 \cdot -b^3$$

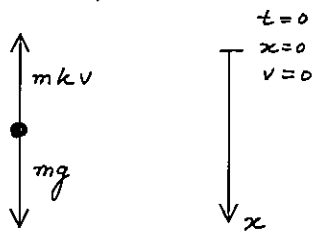
$$a = -2b$$

$$a + 2b = 0$$

Q6 c

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6c(i) Forces on particle



$$ii \quad m\ddot{x} = mg - mkv$$

$$\ddot{x} = k\left(\frac{g}{k} - v\right)$$

as $v \rightarrow \frac{g}{k}$, $\ddot{x} \rightarrow 0$

hence terminal velocity $v = \frac{g}{k}$

$$\therefore \ddot{x} = k(V - v)$$

$$iii) \quad \frac{dv}{dt} = k(V - v)$$

$$-k \frac{dt}{dv} = \frac{-1}{V - v}$$

$$-kt = \ln\{(V - v)A\}, A \text{ constant}$$

$$\left. \begin{matrix} t=0 \\ v=0 \end{matrix} \right\} \Rightarrow A = \frac{1}{V}$$

$$-kt = \ln\left(\frac{V - v}{V}\right)$$

$$t = \frac{1}{k} \ln\left(\frac{V}{V - v}\right)$$

Particle attains 50% of terminal velocity when

$$v = \frac{1}{2}V, \quad t = T = \frac{\ln 2}{k}$$

$$\text{and } x = \frac{V}{2k} (2 \ln 2 - 1)$$

$$\longleftrightarrow v \frac{dv}{dx} = k(V - v)$$

$$\frac{dv}{dx} = k\left(\frac{V - v}{v}\right)$$

$$-k \frac{dx}{dv} = \frac{-v}{V - v}$$

$$= 1 + V \frac{-1}{V - v}$$

$$-kx = v + V \ln\{(V - v)B\}, B \text{ const}$$

$$\left. \begin{matrix} x=0 \\ v=0 \end{matrix} \right\} \Rightarrow B = \frac{1}{V}$$

$$-kx = v + V \ln\left(\frac{V - v}{V}\right)$$

$$x = \frac{1}{k} \left\{ -v + V \ln\left(\frac{V}{V - v}\right) \right\}$$

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ii) But these are alt. angles
 $\therefore AB \parallel DC \Rightarrow ABCD$ is a trap.

$$\begin{aligned} \therefore \angle TAB &= \angle ABT = \alpha \\ &\& \angle ABT = \angle TDC = \alpha \text{ (equal alt. } \angle\text{'s)} \\ \therefore TC &= TD \end{aligned}$$
$$BT = AT \quad (\text{given})$$
$$TC = TD \quad (\text{above})$$
$$\therefore \triangle BTC \equiv \triangle ATD \text{ (S.A.S)}$$
$$\therefore BC = AD \text{ (Corr. sides in Cong. } \Delta s)$$

$\therefore ABCD$ is a trap. with
two equal sides
i.e. isos trap.

$$\therefore \angle VAT = \beta' \text{ (corresponding } \angle\text{'s in cong. } \Delta\text{'s)}$$
$$\therefore \angle UVW = \alpha + \beta \quad \left(\begin{array}{l} \text{EXT. } \angle \text{ cyc. quad} \\ \text{equals int. opp. } \angle \end{array} \right)$$
$$\therefore (\alpha + \beta) + (\alpha + \gamma) = 180^\circ \quad (\text{Co int. \(\angle\)'s are suppl.})$$

But $\angle WXD + (\alpha + \gamma) = 180^\circ$ (opp $\angle$ s
cyc. quad
suppl)

$\therefore UVWX$ is a cyclic quad.
(Ext $\angle$ = int opp $\angle$)

1.2. U, V, W, X are concyclic.

7b

Tuesday, 10 August 2010

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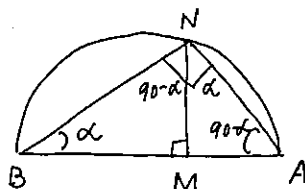
7b(i) consider $\triangle ANM$ and $\triangle NBM$ (similar)

$$\frac{NM}{BM} = \frac{MA}{MN}$$

$$\frac{NM}{q} = \frac{p}{MN}$$

$$NM^2 = pq$$

$$NM = \sqrt{pq} \quad \text{--- (1)}$$



(ii) Now $AB \geq 2NM$ ($\because AB$ is a diameter)
& $AB = p+q$

$$\Rightarrow p+q \geq 2\sqrt{pq}$$

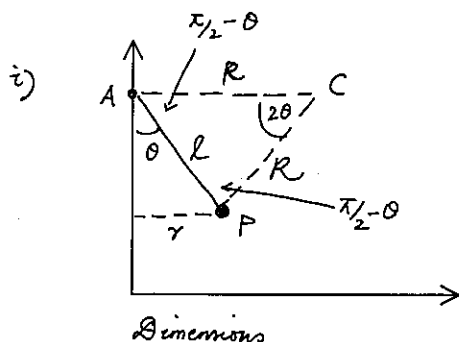
$$\sqrt{pq} \leq \frac{p+q}{2} \quad \text{--- (2)}$$

$$\begin{aligned} \text{(iii)} \quad & \frac{1}{4} (p+q+x+y) \\ &= \frac{1}{2} \cdot \frac{1}{2} (p+q+x+y) \\ &= \frac{1}{2} \left(\frac{p+q}{2} + \frac{x+y}{2} \right) \\ &\geq \frac{1}{2} (\sqrt{pq} + \sqrt{xy}) \quad \text{from (2)} \\ &\geq \sqrt{\sqrt{pq} \sqrt{xy}} \quad \text{from (2)} \\ &= (pqxy)^{1/4} \end{aligned}$$

$$\begin{aligned} \text{(iv)} \quad & \text{using part (iii)} \\ & \frac{l}{m} + \frac{m}{n} + \frac{n}{z} + \frac{z}{l} \geq 4 \left(\frac{l}{m} + \frac{m}{n} + \frac{n}{z} + \frac{z}{l} \right)^{1/4} \\ &= 4(1)^{1/4} \\ \text{i.e.} \quad & \frac{l}{m} + \frac{m}{n} + \frac{n}{z} + \frac{z}{l} \geq 4 \end{aligned}$$

Q8 a

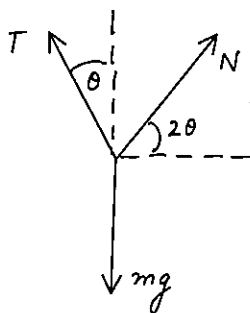
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Force exerted by the surface on the particle is normal to the surface, and hence is directed along the radius towards C.

- ii) ΔACP is isosceles
 $\Rightarrow$ equal angles at A & P
 each $= \frac{\pi}{2} - \theta$
 and angle sum is π
 $\angle ACP = \pi - 2(\frac{\pi}{2} - \theta)$
 $= 2\theta$

iii)



- (iv) Resultant force is directed towards centre of circular path of particle and has magnitude $mr\omega^2$
 $r = l \sin \theta$

Net Vert. Force is zero

$$T \cos \theta + N \sin 2\theta = mg \quad (1)$$

Net Hor. force is $mr\omega^2$ towards y-axis

$$T \sin \theta - N \cos 2\theta = m l \sin \theta \omega^2 \quad (2)$$

(v) $(1) \times \sin \theta - (2) \times \cos \theta$ (eliminate T)

$$\Rightarrow N \cos(2\theta - \theta) = m \sin \theta (g - l \cos \theta \omega^2)$$

$$N \cos \theta = m l \sin \theta \cos \theta \left(\frac{g \sec \theta}{l} - \omega^2 \right)$$

$$N = m l \sin \theta \left(\frac{g \sec \theta}{l} - \omega^2 \right)$$

(vi) $\sin N \geq 0$

$$\frac{g \sec \theta}{l} - \omega^2 \geq 0$$

$$\Rightarrow \omega \leq \sqrt{\frac{g \sec \theta}{l}}$$

Q8 b

Thursday, 12 August 2010

11:13 AM

Q8 (b) i) Let $\tan^{-1}(n+1) = \alpha$

& $\tan^{-1}(n-1) = \beta$

$\Rightarrow \tan \alpha = n+1$ & $\tan \beta = n-1$

Now

$$\tan(\alpha - \beta) = \frac{n+1 - (n-1)}{1 + (n+1)(n-1)} \quad \checkmark$$

$$= \frac{2}{1+n^2-1} = \frac{2}{n^2}$$

$\therefore \alpha - \beta = \tan^{-1} \frac{2}{n^2}$ $\because \alpha - \beta$ & $\tan^{-1} \frac{2}{n^2}$ are $< 90^\circ$

i.e. $\tan^{-1}(n+1) - \tan^{-1}(n-1) = \tan^{-1} \frac{2}{n^2} \quad \checkmark$

ii) $\sum_{r=1}^n \tan^{-1} \frac{2}{r^2} = \tan^{-1} \frac{2}{1^2} + \tan^{-1} \frac{2}{2^2} + \tan^{-1} \frac{2}{3^2} + \dots$

$$+ \dots + \tan^{-1} \frac{2}{(n-2)^2} + \tan^{-1} \frac{2}{(n-1)^2} + \tan^{-1} \frac{2}{n^2}$$

$$= (\tan^{-1} 2 - \tan^{-1} 0) + (\tan^{-1} 3 - \tan^{-1} 1) + (\tan^{-1} 4 - \tan^{-1} 2) + \dots + (\tan^{-1}(n-1) - \tan^{-1}(n-3)) + (\tan^{-1} n - \tan^{-1}(n-2)) + (\tan^{-1}(n+1) - \tan^{-1}(n-1))$$

$$= -\tan^{-1} 0 - \tan^{-1} 1 + \tan^{-1}(n+1) + \tan^{-1} n \quad \checkmark$$

$$= -0 - \frac{\pi}{4} + \pi + \tan^{-1} \left(\frac{n+1+n}{1-(n+1)n} \right)$$

$$= \frac{3\pi}{4} + \tan^{-1} \left(\frac{2n+1}{1-n-n^2} \right) \quad \checkmark$$